

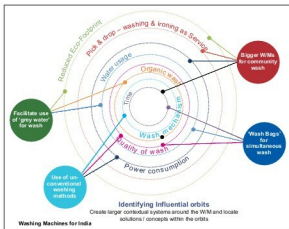
What kind of products should be made

Markets are shifting from the West. India is predicted to become the fifth largest market in the world by 2025 (according to a recent research by the McKinsey Global Institute). While the scope of the Indian market is clear, what exactly this market wants is a question that needs to be answered.

Alok Mittal, managing director of Canaan India, a leading tech venture capitalist in India, says, "Given the mobile revolution in the country, electronics has become mainstream and is not a preserve of the wealthy. This has also brought in a lot of demand for localisation, which represents an opportunity for entrepreneurs and electronics companies. Handsets and telecom equipment are set to become one of the largest categories of imports for the country, and hence local manufacturing has risen up in the country's priority list. All these factors make it attractive for companies to invest in this area. Interestingly, mobile phones are likely to become a platform for adjunct electronics devices, such as those used in remote healthcare and education. We have been seeing start-ups in the healthcare domain as well, like companies that are making sensors, which plug into mobile phones to provide low-cost diagnostics."

According to Malhotra of Idea-farms, "Leveraging the Internet to empower individuals and enable small businesses in semi-urban and rural areas is a foolproof strategy. The increased access to low-cost, high-speed connectivity in rural areas and cheaper access to content in a hyper-local way become powerful change agents in the way information will be delivered, consumed and shared to up the ante in fulfilling basic human needs. Throw in a smartphone and you have a winner. Much of the rural-urban divide is accentuated by a lack of or a lag in information availability. Smartphones, even the cheapest in the market today, will ensure that the consumer is 'always on'."

Srinivas also agrees that the mobile phone is one of the most popular



Designing a washing machine for India (Courtesy: Incubis Consultants)

products in this segment. He adds that smaller-screen television and audio-visual systems, the home-theatre-in-a-box (HTIB) and solar lighting that use renewable energy address the needs of the BoP too, while energy conservation is also a successful theme for this market.

Malhotra spells out a kind of thumb rule. "From my point of view, it makes sense to go from the bottom up; to create a portfolio of products and services to fulfill basic needs before going after the goodies. My list of 'basics' has healthcare, education and finance at a category level; those areas that create better living conditions such as sanitation, clean water, food and energy at the next; and then communication as a means to keeping abreast with the world as well as a channel for feedback, fun and information exchange. Some of the products we saw in the movie '3 Idiots' are pointers to the kind that we need to look at. Not *jugaad* but concerted reuse of available and existing technologies to take these prototypes to the commercialisation stage with the help of interdisciplinary experts and groups," he says.

How do companies make a low-cost product

"Addressing the BoP is neither easy nor cheap. Quite the opposite!" exclaims Malhotra. "There's no intelligence required if all we have to do is to strip a product down to its essentials to reduce quality. I'm also not a big ad-

vocate of inventing niche products to go into these markets. These products must be simple, intuitive, safe and sexy, aka world-class."

Mittal feels that, "In most cases, cost re-engineering is the first important step. This involves understanding the customers at the BoP, and then designing to those specifications. This may involve introducing features (such as ruggedness and long battery life) and eliminating features (such as accelerometers). Since electronics manufacturing is fairly standardised, the functionality drives most of the cost. As you will have noticed, the current examples of BoP products are mostly from the mobile phones' domain—from Nokia's Asha series to Micromax's dual-SIM phones."

Malhotra recalls, "Earlier this year, Amit Gulati, who runs Incubis, invited me to participate in an interactive session to think through design ideas for a low-cost washing machine. The workshop brought out some very interesting and fascinating 'ways of seeing' that completely overturned the engineering/tech/product way of approaching design problems. Did we need to redesign the washing machine (product) under stricter constraints, like the way most people think—start with an existing product, strip it of features, use cheaper materials and processes, reduce quality and make it low-cost. Or did we need to go up a level and reframe the problem itself?"

The answer, he says, lies in the context, which is about culture, geography, climate, behaviour, resources and energy within the boundaries of environmental friendliness and ecology. Everything else is secondary—design, materials, processes, technology and, finally, economic considerations.

When we think of the BoP, we generally tend to put economics first, but according to Malhotra, this is wrong.